

4.1.1 – 4.1.2 Data Sources and Database Connection

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔

CS XII

Chapter 4: Data and Analysis

Lecture 02 of 9

★ Golden

4.1.1 – 4.1.2 Data Sources and Database Connection ★ Golden

LEARN IT

Data sources are where information comes from: school records, websites, sensors, databases.

Connectivity is the conversation between your Python program and that source so you can read and update data.

Five components of a database connection, in the order they actually work:

#	Component	Role
1	Database	Organised store (SQLite = one .db file; MySQL = a server for big apps)
2	Driver	Translator between Python and that DBMS (sqlite3 is built in)
3	Connection object	Opens, manages and closes the pipeline
4	Cursor object	Runs SQL and walks through result rows
5	SQL queries	CREATE, INSERT, SELECT, UPDATE, DELETE

Python → Driver → Connection → Cursor → SQL → Database

EXAM TIP

Write the five names in order and one line each. A common trap is swapping cursor and connection: the connection is the pipe; the cursor is the messenger that travels through it.

اردو میں سمجھیں

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔ کنکشن پائپ ہے، کرسر اس پائپ میں چل کر حکم چلاتا ہے۔

CHECK YOURSELF

1. Which object executes `SELECT *`?
2. Is SQLite a driver, a database, or both?

Answers: 1. The cursor. 2. SQLite is the database; the `sqlite3` module is the driver.

Next lecture: 4.2 Creating a SQLite Database in Python